24B2171- SOC
Physics Informed Neural Netowkrs
Part-C Week 4

Deciding between a conventional FEM approach solver and PiNNs for solving a problem

High-Dimensional PDEs

If the number of dimensions in a problem is high, the number of mesh points increases drastically. For example, say, we want 10 grid points for a unit length on each axis. So, the number of grid points required is 10^d, where d is the dimension of the problem. This increases the computational cost significantly if the problem is approached through FEM, since the size of the matrices and number of equations to be solved parallely is very large. PiNNs are better in this scenario simply because they do not fundamentally rely on grids and meshes to solve the PDE.

Solving Inverse Problems and Data Assimilation

Finite element methods approach problems using forward computations (i.e. given certain parameters like the coefficients and boundary conditions of the PDEs, it proceeds to solve it). However, PiNNs can be used for inverse computation, i.e., given a PDE, the coefficients can be configured to be a trainable parameter along with the solution. The same problem can be approached with FEM too, by repeated computations and some trial and error, but the PiNN approach is way more direct.

Potential Issues with PiNNs

According to the paper written by Grossman et al., PiNNs are also computationally expensive. To achieve the same accuracy as FEM on regular problems, the computational resources required are higher. Also, neural networks suffer from the indeterminate nature of the loss function (we are not dealing with a convex optimisation problem so convergence is not guaranteed). Also, due to spectral bias which was discussed in one of the earlier assignments, PiNNs are in general not very effective at capturing solutions to stiff PDEs, as they fail to capture high frequency changes in the target function. For most standard problems, FEM is optimal, however, for some specific scenarios, such as the two mentioned above, PiNNs may prove to be more effective.

For example, in thermals and fluids engineering, for microheat transfer modelling, Boltzmann Transport Equation has seven dimensions (space, velocity, time), making it difficult to solve using FEM. Similarly, in climate-related modelling, there is a large number of variables. However, the PiNN may fail to capture the variations properly and in a reliable manner; due to which, hybrid models may be used.

Inverse solving is also very popular; for instance, Reynolds number, and some thermal constants can be determined through this approach.